
Technical University of Crete
School of Electrical and Computer Engineering
Course: **Randomized Algorithms**
Exercise 2 (100/1000)
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Maximum Load in n -Balls and n -Bins experiment (20/1000)

Simulate the problem `Balls_and_Bins`(n, n) (for example, $n = 10^k$, $k = 2, 3, 4$). More specifically, assume that you throw n balls into n bins, independently and uniformly at random, and compute the maximum load.

Repeat the experiment K times (for example, $K = 100, 1000$) and plot in the same figure

1. horizontal lines indicating the high-probability lower bound $L = \frac{\ln n}{\ln \ln n}$ and the high-probability upper bound $M = 3 \frac{\ln n}{\ln \ln n}$;
2. the K computed maximum loads.

What do you observe?

Hamiltonian Cycles (80/1000)

Study and implement the modified algorithm for Hamiltonian cycles. More specifically, for large n (for example, $n = 500, 1000$), do the following:

1. Choose $p \in [0, 1]$, with $p \geq \frac{40 \log n}{n}$, and create a random graph $G_{n,p}$.
2. Compute $q \in [0, 1]$, with $p = 2q - q^2$, and create the lists `unused_edges`(v), for every node of the graph v .
3. Run the algorithm in two stages:
 - (a) Run at most $2n \log n$ iterations to compute a Hamiltonian path; if this does not happen, return **FAIL**.

- (b) If the first stage was successful, run at most $n \log n$ iterations to close the cycle; if this does not happen, return **FAIL**.
- 4. If both stages of the algorithm were successful, then
 - (a) Return the number of iterations required in order to (a) compute the Hamiltonian path and (b) close the cycle.
 - (b) Verify the correctness of the Hamiltonian cycle.